

A/D and D/A converters

* Examples where A/D and D/A are used:

1. To monitor the temperature in oven & reduce fuel input.
2. A digital voltmeter is used to measure an analog voltage and display voltage in numerical form.
3. A digital communication system is used to transmit message which are in form of analog electrical signal.

$$* V_o = K(2^{N-1}b_{N-1} + 2^{N-2}b_{N-2} + 2^2b_2 + 2b_1 + b_0)$$

where,

$$I_{N-1} = \frac{V_{N-1}}{R}$$

$$I_{N-2} = \frac{V_{N-2}}{2R}$$

$$I_{N-3} = \frac{V_{N-3}}{2^2 R}$$

where, $V_n = \begin{cases} V(1) & \text{if } b_n = 1 \\ V(0) & \text{if } b_n = 0 \end{cases}$

∴ Output voltage,

$$= \frac{8R}{R} b_3 + \frac{8R}{2R} b_2 + \frac{8R}{2^2 R} b_1 + b_0$$

$$= 8b_3 + 4b_2 + 2b_1 + b_0$$

$$V_o = (-V_R) \left(\frac{R_f}{R} b_{N-1} + \frac{R_f}{2R} b_{N-2} + \frac{R_f}{2^2 R} b_{N-3} + \dots \right)$$

$$K = \frac{R_f}{2^{N-1} R} \cdot V_R$$

$$V_o = \frac{R_f}{2^{N-1} R} \left(2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + 2^{N-3} V_{N-3} + \dots + 2^0 V_0 \right)$$

4 bit $R = 8$
3 bit $R = 4$
5 bit $R = 2$

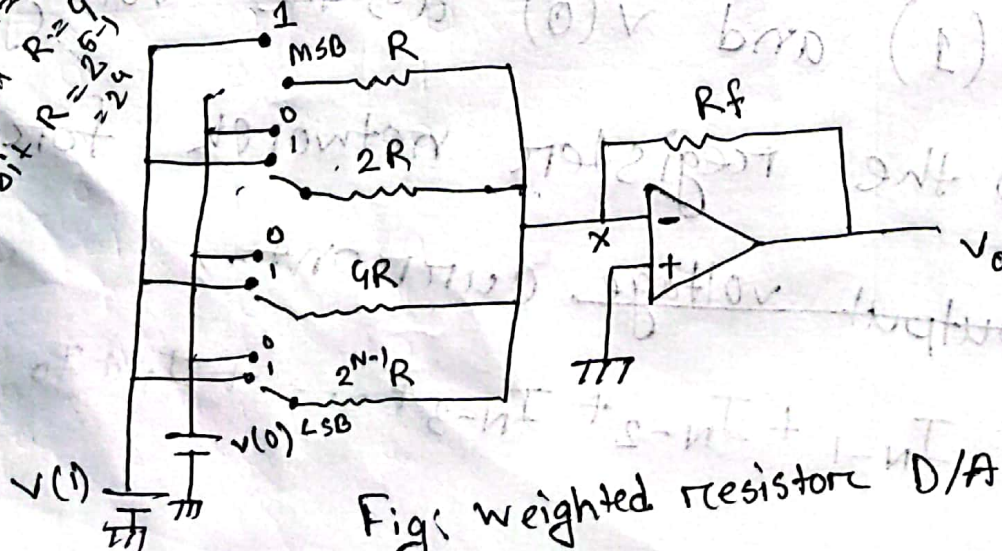


Fig: weighted resistor D/A converter

The offset voltage produced in this circuit,

$$V_{off} = \frac{R_f}{R_{off}} V_{off}$$

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

* Adjust the offset voltage using the circuit such that a digital input of 1000. with this offset obtain the analog output voltage for each of the digital input

Soln

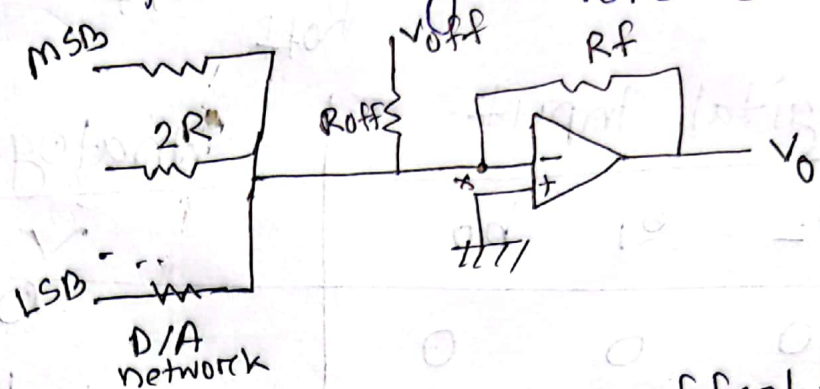


Fig: circuit used to offset the output voltage.

with out offset, the output voltage V_o is 8V for the digital input 1000. The offset must produce a voltage of -8V at the output.

$$-\frac{R_f}{R_{off}} V_{off} = -8V$$

we can use $R_f = R_{off}$, $V_{off} = 1$

The analog voltage are:

Digital input				Analog output
b_3	b_2	b_1	b_0	
0	0	0	0	-8
0	0	0	1	-7
0	0	1	0	-6
0	0	1	1	-5
0	1	0	0	-4
0	1	0	1	-3
0	1	1	0	-2
0	1	1	1	-1
1	0	0	0	0
1	0	0	1	+1
1	0	1	0	+2
1	0	1	1	+3
1	1	0	0	+4
1	1	0	1	+5
1	1	1	0	+6
1	1	1	1	+7

* With the offset used, obtain the analogue output if MSB is complemented before applying to D/A converter.

Soln: If digital input to the D/A converter is $\bar{b}_3 b_2 b_1 b_0$ then analog output voltage are

Digital input				Analog input output
$\bar{b}_3$	b_2	b_1	b_0	
1	0	0	0	-8
1	0	0	1	-7
1	0	1	0	-6
1	0	1	1	-5
1	1	0	0	-4
1	1	0	1	-3
1	1	1	0	-2
1	1	1	1	-1
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
0	0	0	0	8

* What is quantization error?

How it is reduced in A/D converter err?

Soln:

Quantization is the process of analog to digital conversion in which analog voltage intervals are assigned a unique digital value.

The difference between sampled signal to quantized signal is called quantization error.

There are several methods to reduce quantization error:

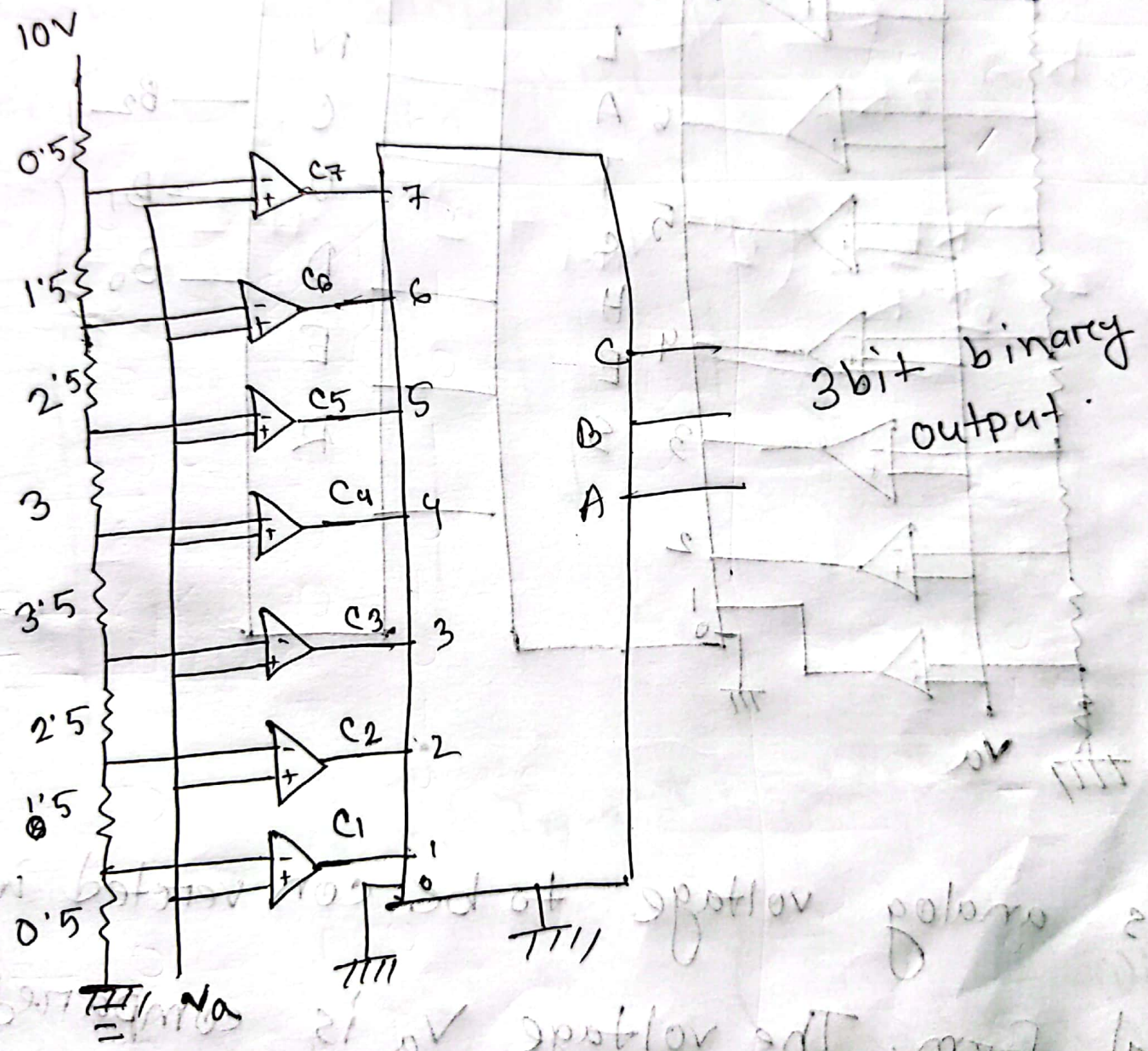
1. Increase the number of bits.
2. User Dither: Introduce a small amount of noise before quantization.
3. Calibration and error correction.
4. Use of sigma: Delta modulation.

~~*Design~~

To reduce quantization error !

Quantization level fix রেখে, কিছুটা

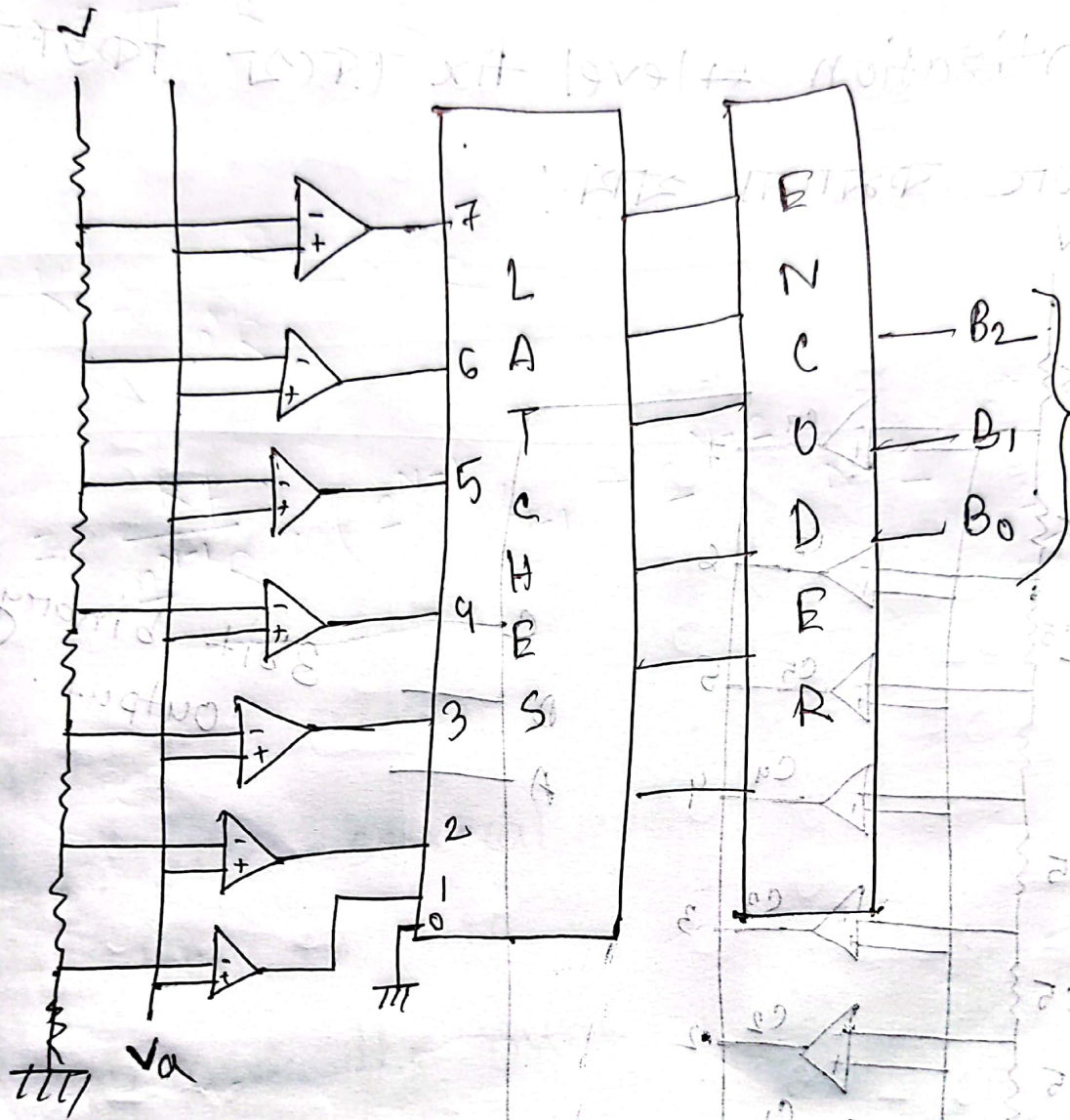
error কমানো যায় !



4 the offset voltage using the circuit

* Design a 3 bit ~~Analog~~ A/D comparator

A/D converter and explain how it works.

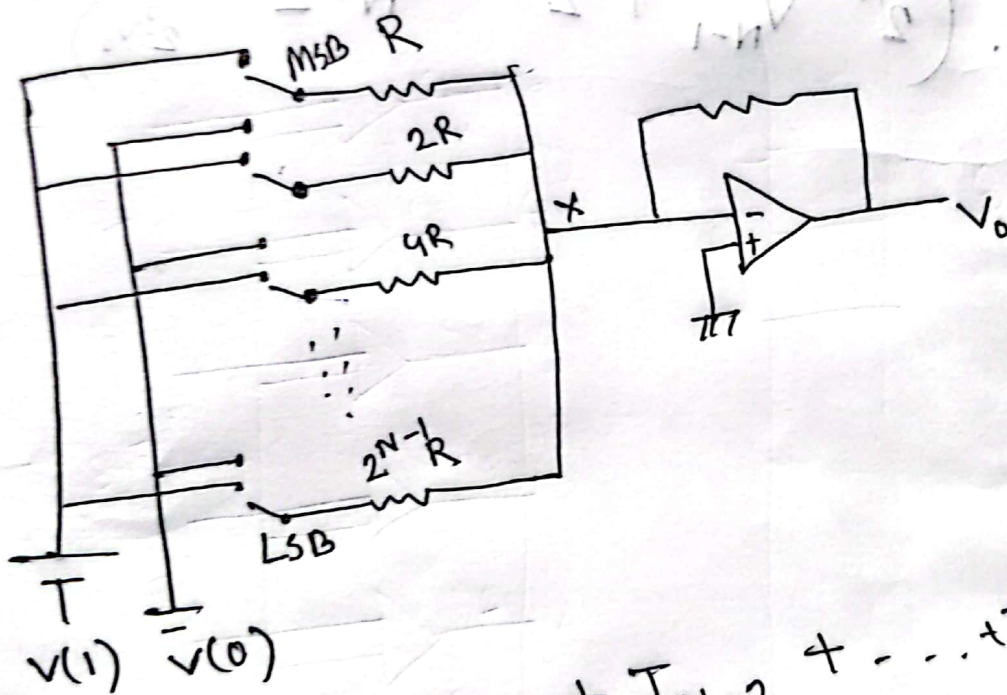


V_a is analog voltage to be converted into digital form. The voltage V_a is compared simultaneously with the reference voltage by using comparators. A 7 bit output is

obtained from the comparators, which is stored in latches. Then 7 bit digital signal is converted to 3 bit output by using a decoder circuit.

* Analyze the output voltage and K factor of the weighted resistor D/A converter

Soln:



$$I_i = I_{N-1} + I_{N-2} + I_{N-3} + \dots + I_1 + I_0$$

where, $I_{N-1} = \frac{V_{N-1}}{R}$

$$I_{N-2} = \frac{V_{N-2}}{R}$$

$$I_{N-3} = \frac{V_{N-3}}{R}$$

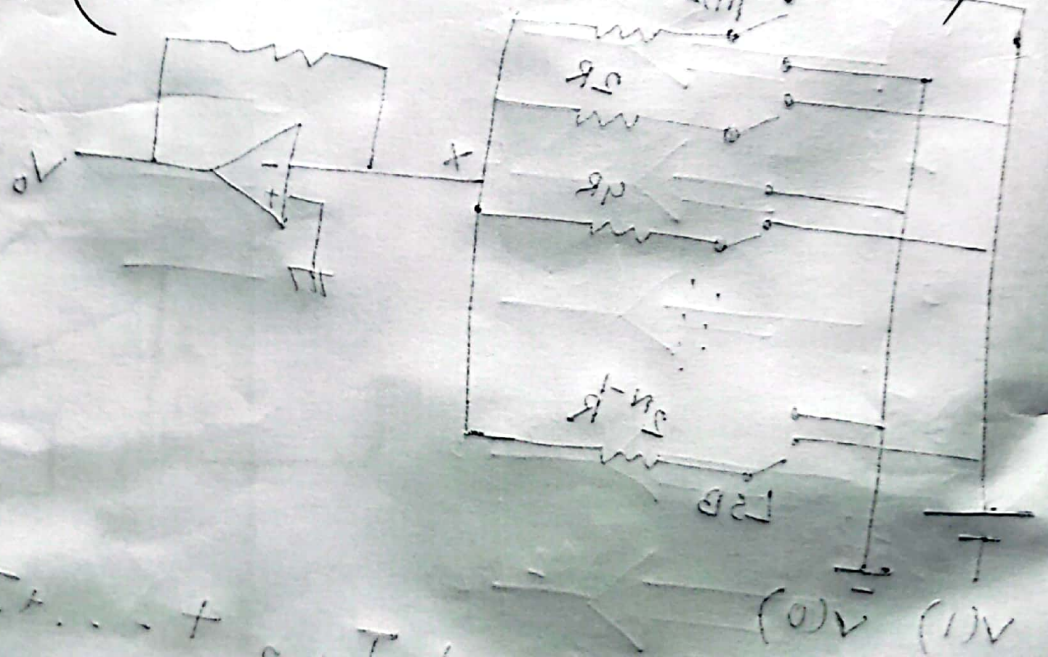
where, $V_n = \begin{cases} V(1) & \text{if } b_n = 1 \\ V(0) & \text{if } b_n = 0 \end{cases}$

Output voltage V_o

$$V_o = (-V_R) \left(\frac{R_F}{R} b_{N-1} + \frac{R_F}{2R} b_{N-2} + \frac{R_F}{2^2 R} b_{N-3} + \dots + \frac{R_F}{2^{N-1} R} b_0 \right)$$

Gain $K = \frac{R_F}{2^{N-1} R} V_R$

$$V_o = \frac{R_F}{2^{N-1} R} \cdot (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \dots + 2^0 V_0)$$



Currents in the ladder network:

$$I_1 = \frac{V(1) - 0}{R} = \frac{V(1)}{R}$$

$$I_2 = \frac{V(2) - 0}{2R} = \frac{V(2)}{2R}$$

$$I_3 = \frac{V(3) - 0}{4R} = \frac{V(3)}{4R}$$

$$\dots$$

$$I_N = \frac{V(N) - 0}{2^{N-1} R} = \frac{V(N)}{2^{N-1} R}$$

~~$V_0 = \frac{2^{N-1} R}{2^{N-1}} \cdot (2^{N-1} - 1)$~~
* Find the analog voltage of 3 bit D/A converter for all possible input

Assume $K=1$.

Soln

We know,

$$V_0 = K \left(2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \dots + 2^0 b_0 \right)$$

So, ~~suppose~~ for 3 bit we will get $= 2^3 = 8$ analog output.

Now

$$V_0 = 2^{3-1} b_{3-1} + 2^{3-2} b_{3-2} + 2^{3-3} b_{3-3}$$

$$= 2^2 b_2 + 2^1 b_1 + 2^0 b_0$$

$$= 4b_2 + 2b_1 + b_0 \quad \text{--- (1)}$$

Now, putting values $b_0 = b_1 = b_2 = 0$,

$$V_0 = 0$$

putting $b_3 = b_2 = 0$ and $b_1 = 1$

$$V_0 = 0 + 0 + 1 = 1$$

putting $b_3 = b_1 = 0$, $b_2 = 1$

$$V_0 = 0 + 2 + 0 = 2$$

Digital input			Analog output
b₃ b ₂	b ₁	b ₀	V
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

* How 4 bit digital output can be obtained from an analog voltage in the range 0 to V

